

College of Science and Computer Engineering

Department of Software Engineering

CCSW 437: Quality of user interaction

Enhancing Usability in



Application

Group Member	ID
Muruj Alshaikh	2210505
Yara Alqarni	2211380
Ayah Mujahid	2211454
Shahad Alqarni	2217309

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First: You define and plan your project

What existing system are you going to test?

Amazon

What users and tasks are you going to test?

Users:

- Frequent Online Shoppers: Individuals who regularly use e-commerce platforms.
 - First-Time Users: New users who have never interacted with Amazon before.
 - Seniors: Older adults who may have different usability needs.
- Amazon Prime Users vs. Non-Prime Users • Mobile vs. Desktop Users.

Tasks:

Homepage:

- Improve the display of recommended products.
- Organize sections better to facilitate navigation.

Search Page:

- Enhance the search algorithm for more accurate results.
- Add advanced filtering options for users.

Product Detail Page:

- Improve image design for larger and clearer views.
- Redesign the review section for easier readability.

Shopping Cart Page:

- Simplify the process for modifying items and managing quantities.
- Provide clear options for completing the purchase or continuing shopping.

Checkout Page:

- Enhance user experience to make the payment process faster and easier.
- Clearly add various payment options.

Account Page:

- Organize personal information and order history better.
- Add features like order tracking in a more visible way.

what are the user experience factors you try to consider?

1. Usability

Ensure users can easily find and access various sections of the site.

2. Accessibility

Make sure the platform is usable for seniors and individuals with disabilities.

3. Performance

Test how quickly pages load, especially on mobile devices. Also, evaluate the responsiveness of the search function and filtering options.

4. Satisfaction

Assess how engaging the content and layout are for different user groups. Also, gather qualitative feedback on users' overall satisfaction with their shopping experience.

Second: Perform usability test for your selected system, to identify some problems with it (in usability test report) and understand how it could be better

What are the user experience metrics and usability scenarios you are going to implement (explain in detail), and why?

No.	Task Title	Task Description	Metrics (can be more than one)	why
1	Explore Homepage	Browse the homepage to view recommended products and navigate through organized sections to find a product of interest.	Task Success Rate, Time on Task, Engagement Metrics, NPS, SUS	Understanding how effectively users navigate the homepage helps identify usability issues and areas for improvement. Metrics like Task Success Rate and Time on Task

reveal how easily users find desired products, while NPS and SUS gauge overall satisfaction and perceived usability

Analyzing how users interact with the search function helps assess the effectiveness of the search algorithm and filtering options. High error rates indicate confusion or difficulty, while Learnability measures how quickly users can adapt to the search features.

2

Use Search Function

Use the search bar to find a specific product and apply advanced filters to narrow down the results for more accurate search outcomes.

Task Success Rate, Time on Task, Error Rate, Learnability, SUS

3

View Product Details

Open a product page, view enlarged and clear product images, and scroll through a redesigned review section for easy readability.

Task Success Rate, Engagement Metrics, Learnability, SUS

Evaluating how users engage with product details is crucial for ensuring they can make informed purchasing decisions. The metrics provide insights into the clarity of product information and

the overall user experience.

4	Modify Cart Items	Add or remove products in the shopping cart and adjust quantities, then decide to either continue shopping or proceed to checkout.	Task Success Rate, Error Rate, Time on Task, Engagement Metrics, SUS	Monitoring this task reveals how intuitive the cart management process is. High error rates may indicate confusion, while engagement metrics help assess user interest and interaction with the cart features.
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5	Complete Checkout	Proceed to the checkout page, select a preferred payment method, and successfully complete the purchase through an improved and faster interface.	Task Success Rate, Error Rate, Time on Task, NPS, SUS	The checkout process is critical for conversion rates. Analyzing this task helps identify barriers to completing purchases, and metrics like Time on Task and NPS provide insights into user satisfaction and efficiency during checkout.
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6	Manage Account	Access the account page to view and manage personal information, browse order history, and track a recent order with improved visibility.	Task Success Rate, Learnability, Retention Rate,	Understanding how users manage their accounts is essential for
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Engagement
Metrics, SUS

enhancing user
loyalty and
satisfaction.
Learnability and
Retention Rate
metrics indicate
how well users
can navigate their
account features,
influencing their
ongoing
engagement with
the platform.

Scenario 1: Explore Homepage

Goal:

Evaluate how easily users can navigate the Amazon homepage and find a product of interest.

Metrics:

- **Task Success Rate** – Did users find a product they're interested in?
- **Time on Task** – How long did it take to explore and choose?
- **Engagement Metrics** – Clicks on recommendations, scroll depth.
- **Satisfaction (SUS/NPS)** – How engaging and personalized was the experience?

Scenario 2: Use Search Function

Goal:

Test how effective the search functionality and filters are in helping users find relevant products.

Metrics:

- **Task Success Rate** – Were relevant products found?
- **Error Rate** – Any wrong or irrelevant clicks?
- **Time on Task** – How long did it take to filter to desired results?
- **Learnability** – How easy was it to use search and filters?
- **SUS Score** – Usability feedback.

Scenario 3: View Product Details

Goal:

Determine how effectively product information and visuals help in decision-making.

Metrics:

- **Task Success Rate** – Were users able to gather enough info?
- **Engagement Metrics** – Time spent on product images and reviews.
- **Learnability** – Were elements intuitive (zooming images, reading specs)?
- **SUS Score** – Was the product detail page helpful and clear?

Scenario 4: Modify Cart Items

Goal:

Assess how easily users can manage their shopping cart items.

Metrics:

- **Task Success Rate** – Were cart changes made successfully?
- **Error Rate** – Any incorrect items removed/added?
- **Time on Task** – Efficiency in managing cart.
- **Engagement Metrics** – Clicks, use of “save for later” or quantity selector.
- **SUS Score** – Satisfaction with cart interface.

Scenario 5: Complete Checkout

Goal:

Evaluate the speed, ease, and clarity of the checkout process.

Metrics:

- **Task Success Rate** – Was the checkout completed?
- **Error Rate** – Any mistakes with address/payment?
- **Time on Task** – How long to finalize the order?
- **NPS** – Would users recommend the checkout experience?
- **SUS Score** – Was it smooth and stress-free?

Scenario 6: Manage Account

Goal:

Test how intuitive and effective the account page is for viewing and managing personal info and orders.

Metrics:

- **Task Success Rate** – Were all tasks completed?
- **Learnability** – Was it easy to find the right settings?
- **Retention Rate** – Would users return to manage their accounts again?

- **Engagement Metrics** – Number of clicks, time spent.
- **SUS Score** – Was the account dashboard clear and helpful?

What are the data collection methods you are going to implement (explain in detail), type of data measure, and why?

Data Collection Methods

- Observation: Observe users as they interact with the Amazon website, paying attention to how they navigate through product categories, use filters, or proceed through checkout. Note where they experience confusion, hesitation, or dissatisfaction.
- Surveys/Questionnaires: Distribute surveys to collect qualitative feedback on users' experiences with Amazon. Focus on areas such as ease of finding products, checkout process, interface, and overall satisfaction.
- User Interviews: Conduct short interviews to gather in-depth insights into user expectations, frustrations, and perceptions of the Amazon site design and functionality.

Types of Data

Quantitative Data:

- Task Success Rate: Percentage of users who successfully complete key tasks (e.g., searching for a product, completing a purchase).
- Time on Task: Time taken to perform specific tasks like locating a product or completing checkout.
- Error Rate: Number of mistakes made during tasks (e.g., incorrect use of filters, failed attempts to add items to cart).

Qualitative Data:

User Feedback: Open-ended comments about their experience using Amazon.

Satisfaction Ratings: Users' self-reported levels of satisfaction with features such as product search, product detail pages, or the shopping cart.

What are the data analysis methods you are going to implement (explain in detail), and why?

1. Task Completion Rate

- Measures: % of users who successfully complete a given task.

-Use: Shows how effective the Amazon website is in helping users achieve goals.

2. Time on Task

-Measures: How long users take to complete a task.

-Use: Indicates how efficient the site is. Shorter time = more intuitive experience.

3. Error Rate

-Measures: Number of mistakes users make (e.g., clicking wrong buttons, backtracking, failed form submissions).

-Use: Helps identify areas of poor usability or confusing design.

4. Clickstream Analysis

-Measures: Sequence of pages or actions a user takes.

-Use: Tracks user journeys — helps understand user flow and where they drop off.

5. Survey & Feedback Analysis

-Analyses: Responses from questionnaires or user comments.

-Use: Pulls out themes, satisfaction levels, or common pain points.

What is the data presentation forms you are going to use (explain in detail), and why?

1. Bar Graphs

Why it's a good choice:

Bar graphs are simple and easy to understand. They help compare results for different tasks like searching for products, checking out, or using the shopping cart. We can use a bar graph to show how many users were successful in each task or how long each task took.

They're great for showing:

- **Task success rate**
- **Time on task**
- **Satisfaction scores (like SUS or NPS)**

2. Stacked Bar Graphs

Why it's a good choice:

Stacked bar graphs are useful when you want to show a total amount and also break it into smaller parts. We can show how many errors happened during checkout and then break it down by type of user

They're great for showing:

- **Error rates by user group**
- **Engagement like clicks or time spent by platform (mobile vs. desktop)**

What is the issue you are going to improve (explain in detail), and how?

The issue we are going to improve is the **overall user experience on Amazon**, especially focusing on **usability, accessibility, performance, and satisfaction** across different parts of the platform like the homepage, search, product details, shopping cart, checkout, and account pages.

Why is this important?

Amazon is a large and widely-used platform, but not all users have the same experience. Some users (like seniors or first-time users) may find it hard to navigate, while others might struggle with the layout, finding products, or completing purchases. If users can't complete their tasks easily, they may leave the site or feel frustrated, which affects Amazon's success.

Specific Problems We Want to Improve:

- 1. Homepage Organization**
 - We want to improve the layout and how recommended products are displayed to help users find what they want more easily.
- 2. Search and Filters**
 - We aim to improve the search algorithm and add better filtering options so users can find products faster.
- 3. Product Detail Page**
 - We want to improve image zooming and redesign the review section for better readability.
- 4. Shopping Cart**
 - We plan to make cart management more user-friendly and provide clear options for continuing shopping or checking out.
- 5. Checkout Process**

- We will work on simplifying the steps, improving the layout, and showing more payment options clearly.
6. **Account Page**
- We want to organize the page better and make features like order tracking more visible.

How are we going to improve these issues?

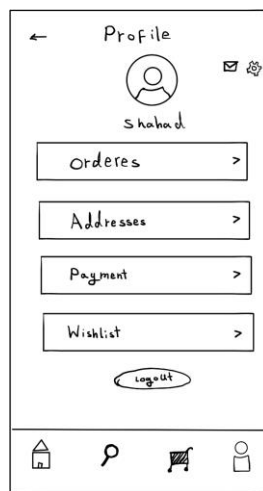
1. **Testing with Different User Groups**
 - We will test with various types of users, including frequent shoppers, first-time users, seniors, Prime vs. Non-Prime users, and mobile vs. desktop users. This will help us understand different needs and challenges.
2. **Using Real Tasks and Scenarios**
 - We created 6 user scenarios like “Explore Homepage” and “Complete Checkout” and will observe how users perform these tasks.
 - We will use metrics such as:
 - **Task Success Rate**
 - **Time on Task**
 - **Error Rate**
 - **SUS (System Usability Scale) Score**
 - **NPS (Net Promoter Score)**
 - **Engagement and Learnability Metrics**
3. **Presenting the Results Visually**
 - We will use **bar graphs** to compare performance across tasks and show results like how long tasks took or how satisfied users were.
 - We’ll also use **stacked bar graphs** to break down data by user type (like mobile vs. desktop or Prime vs. Non-Prime), making it easy to see patterns or differences.

Finally: Analyse your concerns regarding the issues and/or redesign and otherwise innovate and augment the selected version

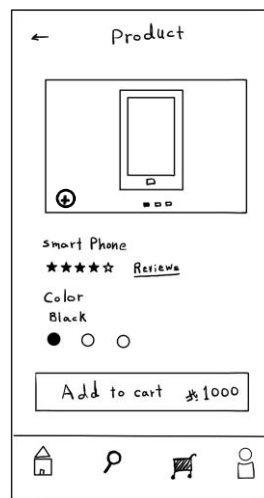
Driven by the usability problems you identified in the test, what is the wireframe (prototype) are you going to develop to improve the experience of the existing system?

We chose to improve the experience of this six pages

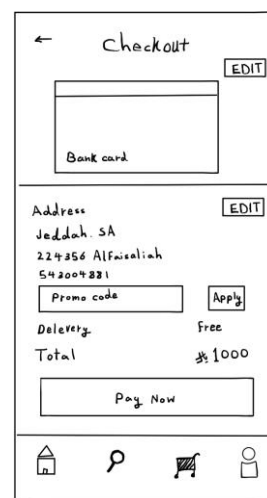
- Low fidelity:



Profile page



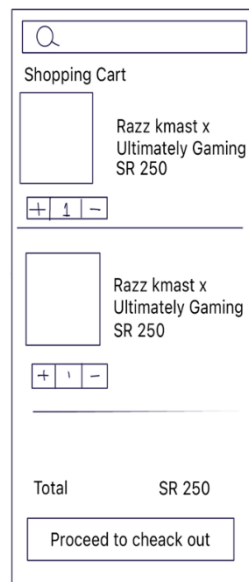
Product details



Checkout page



Homepage



Shopping cart



Search

Complete the high-fidelity version of your redesign (either as interactive wireframe or html/css or interactive wireframe)

We redesign the Amazon Application by using Figma

The Link:

<https://www.figma.com/proto/2iaeYOdH6xVDC1UvanTD39/Amazon?node-id=0-1&t=caugF0slY8tjMX76-1>

- High fidelity

